

ELECTIVE COURSE-3.1

URA416: HUMAN MACHINE INTERFACE				
	L	T	P	Cr
	2	2	0	3.0
Course Objective: This course is designed to learn the designing and develop effective and user-friendly HMI systems for a variety of domains, including automotive, industrial, consumer electronics, medical, and aerospace.				
Syllabus Introduction to HMI and domains, Automotive electronics: Automotive, Industrial, CE, Medical, Aero, ECUs within car and their functionalities. Interaction between ECUs. (CAN, LIN, Most, FlexRay, Ethernet etc) Car Multimedia: Instrument Cluster, In Vehicle Infotainment, Professional System, Rear Seat Entertainment, Evolution, Overview, H/W, S/W and mechanics introduction Automotive HMI: GenIVI, Automotive Safety - Processes, General introduction to HMI, Evolution of HMI in car, HMI for Car Multimedia (IS, IVI, PS, RSE) - same as module 2, GUI Tools (GTK, Qt, WinCE, QNX, Clutter, HTML5 etc), VUI, HMI Architecture & Concepts, HMI Sub Components (Widgets, Framework, statemachine) UX and Guidelines: Introduction to UX design - stages, theory, Design thinking, UX Study, Interaction concepts, Graphic design (with introduction to tools - Adobe Photoshop, Adobe XD, Blender, GIMP ..etc) - Asset Design - Overview only, Guidelines and norms, 2D/3D rendering, OpenGL, OSG etc, GPU architectures, Shader Programming App Development: Qt, Unity, Android, iOS, Hybrid - UI, PWA, HTML CSS Javascript (Front End Frameworks), IAR or yakindo, OpenCV Trends and advanced topics (Awareness): Voice, Gesture, Vision, sensor based UI controls, Haptics, New technologies (eyegaze, gesture, dual display etc.), SPI - android auto, car play., Smart				

Approved in 109th meeting of the Senate held on March 16, 2023. Revised in 112th and 113th meeting of the Senate held on March 11 and September 7, 2024, respectively. Further, revision Approved in 1st meeting of Academic Council held on September 11, 2025.

City and Public Transport, ride sharing, personal, Virtual Reality , Augmented Reality and Mixed Reality, UI Analytics (Usage patterns, ..), Debugging, Performance Profiling

Course Learning Objectives (CLO)

The students will be able to:

1. understand the principles of HMI design and development.
2. apply user experience (UX) design principles to HMI development.
3. develop HMI applications for a variety of domains, including automotive, industrial, consumer electronics, medical, and aerospace.
4. apply new HMI technologies to the development of innovative and user-friendly HMI systems.

Text Books

1. Shuo Gao, Shuo Yan, Hang Zhao, Arokia Nathan, "Touch-Based Human-Machine Interaction: Principles and Applications", Springer Nature Switzerland AG; 1st edition, 2021.
2. Robert Wells, "Unity 2020 By Example: A project-based guide to building 2D, 3D, augmented reality, and virtual reality games from scratch", Packt Publishing Limited, 2020.

Reference Books & Websites

1. Ryan Cohen, Tao Wang, "GUI Design for Android Apps", Apress, Berkeley, CA, 2014.

Evaluation Scheme:

Sl. No	Evaluation Elements	Weightage (%)
1	MST	25

Approved in 109th meeting of the Senate held on March 16, 2023. Revised in 112th and 113th meeting of the Senate held on March 11 and September 7, 2024, respectively. Further, revision Approved in 1st meeting of Academic Council held on September 11, 2025.